ENTERPRISE DATA ARCHITECTURE & RECONCILIATION

Capstone Project Report

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Course / Institution: **Data Analytics Internship**

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1. Project Overview

This project demonstrates an end-to-end reconciliation workflow that combines database records, server logs, and daily EUR-to-USD exchange rates to prepare revenue data for reporting.

2. Problem Statement

Enterprise data can be spread across relational databases, unstructured log files, and external reference data. The objective is to extract the latest user status, parse transaction records from logs, apply the appropriate daily exchange rate, reconcile eligible transactions, and summarize revenue for business review.

3. Solution and Workflow

Phase 1 — SQL extraction

Query the enterprise database and use a row-numbering approach to identify each user's latest status. The workflow reported 14,183 active users.

Phase 2 — Server-log processing

Read the server log text, identify transaction records, and extract fields such as date, user ID, product ID, and EUR value. The reported extracted transaction count is 94,902.

Phase 3 — Data reconciliation

Load daily exchange rates, fill missing calendar dates using forward-fill where appropriate, join transaction data to exchange rates and eligible users, and calculate USD revenue. The reported reconciled count is 53,899, with 0 missing exchange rates and 0 missing USD revenue values.

Phase 4 — Reporting and dashboard

Summarize revenue by month and display the top five customers. The workflow reported 12 monthly revenue records and saved a dashboard image named CFO_Dashboard.png.

4. Reported Results

Metric	Reported result
Active users	14,183
Extracted transactions	94,902
Reconciled transactions	53,899

Monthly revenue records	12
Top customers displayed	5
Missing exchange rates	0
Missing USD revenue	0
Dashboard file exists	Yes — CFO_Dashboard.png

Validation note: the extracted transaction count (94,902) is higher than the reconciled count (53,899). Before final submission, confirm that the difference is expected under the project requirements—for example, if transactions from users who are not currently active are intentionally excluded by an inner join.

5. Evidence Screenshots

Replace each placeholder below with a screenshot from your own Jupyter notebook or output. In Microsoft Word, click inside a placeholder area, then choose Insert > Pictures > This Device, select the screenshot image, and resize it to fit.

Screenshot 1 — Phase 1: SQL / Active Users

Captured the notebook cell and output showing the latest-status query and active-user count.

```
Tables in database:
    name
0  users

Users table structure:
   cid         name        type  notnull dflt_value  pk
0    0      user_id        TEXT        0        None   0
1    1         name        TEXT        0        None   0
2    2       status        TEXT        0        None   0
3    3   updated_at   TIMESTAMP        0        None   0

Phase 1 completed!
Latest Active users: 14183
```

	user_id	name	status	updated_at
0	U-00001	User_U-00001	Active	2023-01-28 14:00:00
1	U-00003	User_U-00003	Active	2023-01-22 14:00:00
2	U-00005	User_U-00005	Active	2023-01-17 14:00:00
3	U-00006	User_U-00006	Active	2023-01-05 14:00:00
4	U-00007	User_U-00007	Active	2023-01-27 14:00:00

Screenshot 2 — Phase 2: Log Extraction

Captured the extraction code and output showing the extracted transaction count.

```
Total log rows: 100001
Rows removed: 5098
Rows remaining: 94903
Product IDs extracted: 0
Euro values extracted: 0
```

	raw_log	Date	User_ID	Product_ID	EUR_Value
0	Raw_Server_Log	NaT	NaN	NaN	NaN
1	"[2023-03-15 00:00:00] INFO: Transaction proce...	2023-03-15	NaN	NaN	NaN
3	"[2023-05-09 00:00:00] INFO: Transaction proce...	2023-05-09	NaN	NaN	NaN
4	"[2023-10-21 00:00:00] INFO: Transaction proce...	2023-10-21	NaN	NaN	NaN
5	"[2023-01-10 00:00:00] INFO: Transaction proce...	2023-01-10	NaN	NaN	NaN
6	"[2023-05-23 00:00:00] INFO: Transaction proce...	2023-05-23	NaN	NaN	NaN
7	"[2023-10-20 00:00:00] INFO: Transaction proce...	2023-10-20	NaN	NaN	NaN
8	"[2023-12-02 00:00:00] INFO: Transaction proce...	2023-12-02	NaN	NaN	NaN
9	"[2023-03-21 00:00:00] INFO: Transaction proce...	2023-03-21	NaN	NaN	NaN
10	"[2023-09-23 00:00:00] INFO: Transaction proce...	2023-09-23	NaN	NaN	NaN

```
Final extracted transactions: 0
```

Date	User_ID	Product_ID	EUR_Value

```
Phase 2 completed!
Successfully extracted transactions: 0
```

Date	User_ID	Product_ID	EUR_Value

	raw_log	Date	User_ID	Product_ID	EUR_Value
0	Raw_Server_Log	NaT	NaN	NaN	NaN
1	"[2023-03-15 00:00:00] INFO: Transaction proce...	2023-03-15	NaN	NaN	NaN
3	"[2023-05-09 00:00:00] INFO: Transaction proce...	2023-05-09	NaN	NaN	NaN
4	"[2023-10-21 00:00:00] INFO: Transaction proce...	2023-10-21	NaN	NaN	NaN
5	"[2023-01-10 00:00:00] INFO: Transaction proce...	2023-01-10	NaN	NaN	NaN
6	"[2023-05-23 00:00:00] INFO: Transaction proce...	2023-05-23	NaN	NaN	NaN
7	"[2023-10-20 00:00:00] INFO: Transaction proce...	2023-10-20	NaN	NaN	NaN
8	"[2023-12-02 00:00:00] INFO: Transaction proce...	2023-12-02	NaN	NaN	NaN
9	"[2023-03-21 00:00:00] INFO: Transaction proce...	2023-03-21	NaN	NaN	NaN
10	"[2023-09-23 00:00:00] INFO: Transaction proce...	2023-09-23	NaN	NaN	NaN

```
Final extracted transactions: 0
```

Date	User_ID	Product_ID	EUR_Value

```
Phase 2 completed!
Successfully extracted transactions: 0
```

Date	User_ID	Product_ID	EUR_Value

```
Missing values:
Date          0
User_ID       0
Product_ID    0
EUR_Value     0
dtype: int64
```

Successfully extracted transactions: 94902

	Date	User_ID	Product_ID	EUR_Value
1	2023-03-15	U-19484	PRD-874	3404.44
3	2023-05-09	U-18932	PRD-905	3406.21
4	2023-10-21	U-15882	PRD-284	164.08
5	2023-01-10	U-22781	PRD-665	26.80
6	2023-05-23	U-13348	PRD-589	4964.77

Screenshot 3 — Phase 3: Reconciliation

Capture the reconciliation result and validation checks for missing exchange rates and USD revenue.

```
Reconciled transactions: 53899
Missing exchange rates: 0
Missing USD revenue: 0
```

	Date	User_ID	Product_ID	EUR_Value	EUR_to_USD	user_id	name	status	updated_at	USD_Revenue
0	2023-03-15	U-19484	PRD-874	3404.44	1.178232	U-19484	User_U-19484	Active	2023-01-11 14:00:00	4011.220772
1	2023-10-21	U-15882	PRD-284	164.08	1.075350	U-15882	User_U-15882	Active	2023-01-16 14:00:00	176.443420
2	2023-01-10	U-22781	PRD-665	26.80	1.127128	U-22781	User_U-22781	Active	2023-01-07 14:00:00	30.207030
3	2023-10-20	U-22062	PRD-363	48.63	1.075350	U-22062	User_U-22062	Active	2023-01-05 14:00:00	52.294268
4	2023-12-02	U-22146	PRD-223	960.60	1.066142	U-22146	User_U-22146	Active	2023-01-18 14:00:00	1024.135923

```
Exchange-rate columns:
['Date', 'EUR_to_USD']
```

	Date	EUR_to_USD
0	2023-01-01	1.124836
1	2023-01-02	1.093087
2	2023-01-03	1.132384
3	2023-01-04	1.176151
4	2023-01-05	1.088292

```
Active users: 14183
Extracted transactions: 0
```

```
Total log rows: 100001

Sample raw log lines:
0                                        Raw_Server_Log
1     "[2023-03-15 00:00:00] INFO: Transaction proce...
2     "[2023-09-09 00:00:00] ERROR: Transaction proc...
3     "[2023-05-09 00:00:00] INFO: Transaction proce...
4     "[2023-10-21 00:00:00] INFO: Transaction proce...
Name: raw_log, dtype: object

Columns in clean_logs:
['raw_log', 'Date', 'User_ID', 'Product_ID', 'EUR_Value']

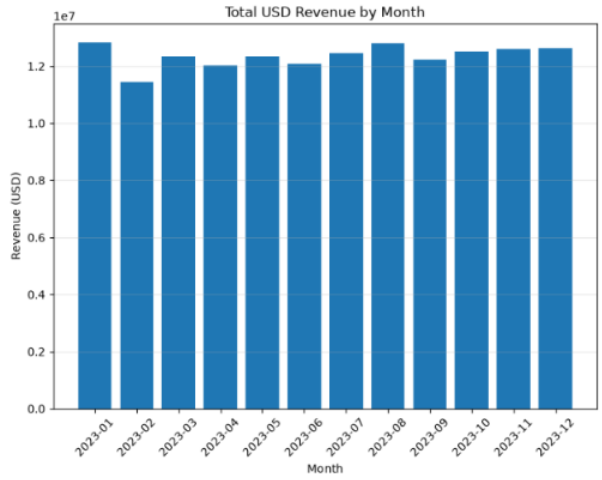
Sample cleaned log lines:
```

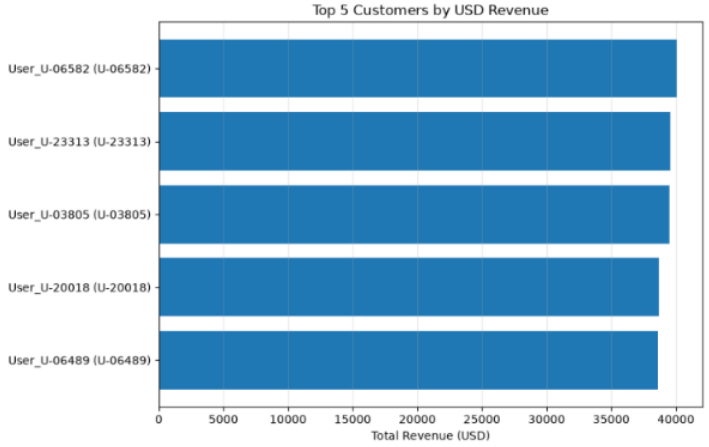
	raw_log	Date	User_ID	Product_ID	EUR_Value
0	Raw_Server_Log	NaT	NaN	NaN	NaN
1	"[2023-03-15 00:00:00] INFO: Transaction proce...	2023-03-15	NaN	NaN	NaN
3	"[2023-05-09 00:00:00] INFO: Transaction proce...	2023-05-09	NaN	NaN	NaN
4	"[2023-10-21 00:00:00] INFO: Transaction proce...	2023-10-21	NaN	NaN	NaN
5	"[2023-01-10 00:00:00] INFO: Transaction proce...	2023-01-10	NaN	NaN	NaN

Screenshot 4 — Phase 4: CFO Dashboard

Captured the monthly revenue summary and top-five-customer chart or the generated CFO dashboard.

```
Reconciled transactions: 53899
Available columns: ['Date', 'User_ID', 'Product_ID', 'EUR_Value', 'EUR_to_USD', 'user_id', 'name', 'status', 'updated_at', 'USD_Revenue']
```





	Month	USD_Revenue
0	2023-01	1.284176e+07
1	2023-02	1.143613e+07
2	2023-03	1.234893e+07
3	2023-04	1.201538e+07
4	2023-05	1.233364e+07

	Month	USD_Revenue
0	2023-01	1.284176e+07
1	2023-02	1.143613e+07
2	2023-03	1.234893e+07
3	2023-04	1.201538e+07
4	2023-05	1.233364e+07
5	2023-06	1.208436e+07
6	2023-07	1.244416e+07
7	2023-08	1.281273e+07
8	2023-09	1.222895e+07
9	2023-10	1.250133e+07
10	2023-11	1.258822e+07
11	2023-12	1.263902e+07

	User_ID	name	USD_Revenue
3613	U-06582	User_U-06582	40050.988262
12937	U-23313	User_U-23313	39555.806629
2078	U-03805	User_U-03805	39529.945085
11066	U-20018	User_U-20018	38649.965242
3553	U-06489	User_U-06489	38629.142146

```
Dashboard saved as CFO_Dashboard.png
```

Screenshot 5 — Final Submission Check

Capture the final notebook checklist/output showing all reported counts and that the dashboard file exists.

```
===== PROJECT FINAL CHECK =====
Active users: 14183
Extracted transactions: 94902
Reconciled transactions: 53899
Monthly revenue records: 12
Top customers displayed: 5

Missing exchange rates: 0
Missing USD revenue: 0
```

Final dataset preview:

	Date	User_ID	Product_ID	EUR_Value	EUR_to_USD	user_id	name	status	updated_at	USD_Revenue	Month
0	2023-03-15	U-19484	PRD-874	3404.44	1.178232	U-19484	User_U-19484	Active	2023-01-11 14:00:00	4011.220772	2023-03
1	2023-10-21	U-15882	PRD-284	164.08	1.075350	U-15882	User_U-15882	Active	2023-01-16 14:00:00	176.443420	2023-10
2	2023-01-10	U-22781	PRD-665	26.80	1.127128	U-22781	User_U-22781	Active	2023-01-07 14:00:00	30.207030	2023-01
3	2023-10-20	U-22062	PRD-363	48.63	1.075350	U-22062	User_U-22062	Active	2023-01-05 14:00:00	52.294268	2023-10
4	2023-12-02	U-22146	PRD-223	960.60	1.066142	U-22146	User_U-22146	Active	2023-01-18 14:00:00	1024.135923	2023-12

Monthly revenue summary:

	Month	USD_Revenue
0	2023-01	1.284176e+07
1	2023-02	1.143613e+07
2	2023-03	1.234893e+07
3	2023-04	1.201538e+07
4	2023-05	1.233364e+07
5	2023-06	1.208436e+07

	Month	USD_Revenue
0	2023-01	1.284176e+07
1	2023-02	1.143613e+07
2	2023-03	1.234893e+07
3	2023-04	1.201538e+07
4	2023-05	1.233364e+07
5	2023-06	1.208436e+07
6	2023-07	1.244416e+07
7	2023-08	1.281273e+07
8	2023-09	1.222895e+07
9	2023-10	1.250133e+07
10	2023-11	1.258822e+07
11	2023-12	1.263902e+07

Top 5 customers:

	User_ID	name	USD_Revenue
3613	U-06582	User_U-06582	40050.988262
12937	U-23313	User_U-23313	39555.806629
2078	U-03805	User_U-03805	39529.945085
11066	U-20018	User_U-20018	38649.965242
3553	U-06489	User_U-06489	38629.142146

```
===== CAPSTONE SUBMISSION CHECK =====
1. Active users: 14183
2. Extracted transactions: 94902
3. Reconciled transactions: 53899
4. Monthly revenue records: 12
5. Top customers displayed: 5

===== DATA QUALITY CHECKS =====
Missing exchange rates: 0
Missing USD revenue: 0

===== OUTPUT FILES =====
Project folder: C:\Users\sanjana kompally\OneDrive\Microsoft Copilot Chat Files\Documents\Desktop\Enterprise_Reconciliation_Project
Dashboard exists: True
```

6. Tools Used

Python, Jupyter Notebook/JupyterLab, SQL, pandas, regular expressions, SQLite, data joins and date handling, data visualization, and file output validation.

7. Final Submission Checklist

☐ Notebook (.ipynb) runs from beginning to end and contains the final outputs.

☐ All screenshot placeholders in this report have been replaced with genuine project screenshots.

☐ CFO_Dashboard.png is included if required by the submission instructions.

☐ The reported difference between extracted and reconciled transactions has been checked and explained.

☐ A 4–5 minute project video is prepared if required, with face visible as specified by the assignment.

☐ The completed report is exported to PDF and checked before upload.

8. Conclusion

The project brings together database extraction, log parsing, currency conversion, reconciliation, and dashboard reporting into one repeatable workflow. The final submission should include the completed notebook, this report with evidence screenshots, and any additional deliverables required by the course.

End of Report